

## Scalability & Future Plan

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|----------------------|----------------------------------------------------------|
| <b>Date</b>          | 5 July 2024                                              |
| <b>Team ID</b>       | SWTID-2026-4860                                          |
| <b>Project Name</b>  | <b>EduGenie Google Gemini Powered Learning Assistant</b> |
| <b>Maximum Marks</b> | 5 Mark                                                   |

### Scalability & Future Plan:

This document outlines how the **EduGenie – Google Gemini Powered Learning Assistant** can be scaled to support a larger number of users, increased data volumes, and additional AI-powered features in the future. It also presents the project's roadmap for continuous improvement and long-term growth.

### Current System Limitations:

| S.No | Limitation                                                | Impact                                                                                     | Priority to Address<br>(High / Medium / Low) |
|------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------|
| 1    | Limited concurrent user support                           | System performance may decrease during peak usage.                                         | High                                         |
| 2    | Dependence on Google Gemini API and internet connectivity | AI features become unavailable if the API or network is inaccessible.                      | High                                         |
| 3    | Limited personalization and analytics                     | Learning recommendations are basic and do not fully adapt to individual learning patterns. | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                            | Proposed Upgrade / Solution                                                                              |
|------|--------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a limited number of simultaneous users.         | Deploy on cloud infrastructure with load balancing and auto-scaling to support thousands of users.       |
| 2    | Data Storage       | Stores basic user and chat data.                         | Use scalable cloud databases with automated backup, replication, and efficient data management.          |
| 3    | Performance        | Response time depends on API calls and server resources. | Implement caching, asynchronous processing, and optimize API requests to improve response time.          |
| 4    | Security           | Basic authentication and secure API communication.       | Add role-based access control, multi-factor authentication, encryption, and regular security monitoring. |

### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                 | Target Timeline | Expected Impact                                                         |
|---------|-------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------|
| Phase 2 | Personalized learning dashboard, progress tracking, and multilingual support. | 3–6 Months      | Improves user engagement and provides a customized learning experience. |

|         |                                                                                                 |              |                                                                                          |
|---------|-------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------|
| Phase 3 | Voice-based interaction, AI tutor with speech recognition, and mobile application support.      | 6–12 Months  | Enhances accessibility and enables learning anytime, anywhere.                           |
| Phase 4 | LMS integration, collaborative learning, advanced analytics, and offline learning capabilities. | 12–18 Months | Makes EduGenie suitable for schools, colleges, and large-scale educational institutions. |